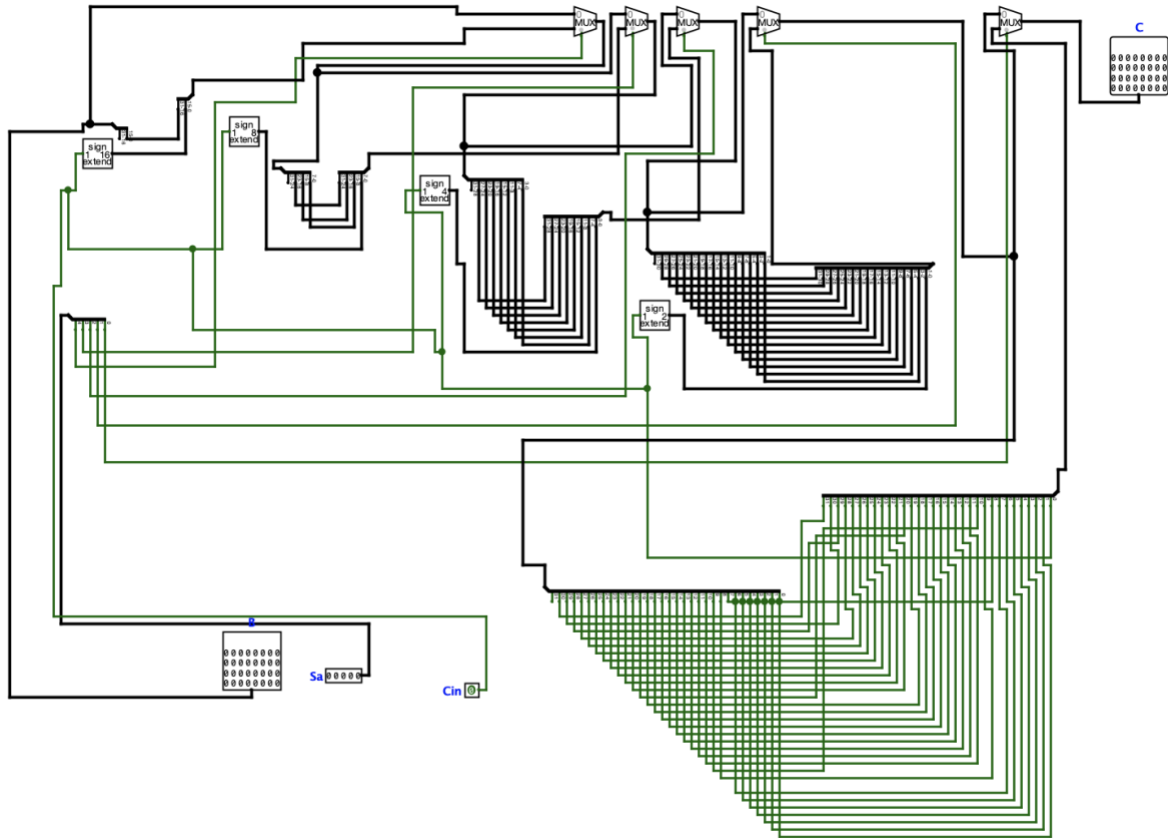
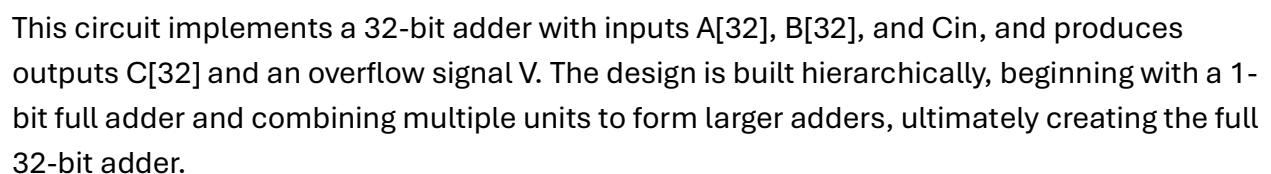


[illegible]



This circuit performs a 32-bit left shift operation. It uses inputs B [32], Sa[5], and Cin to generate the output C [32]. The value in B is shifted left by a number of positions defined by Sa, ranging from 0 to 31. Bits that move past the left end are dropped, and the vacant positions on the right are filled with the value of Cin.

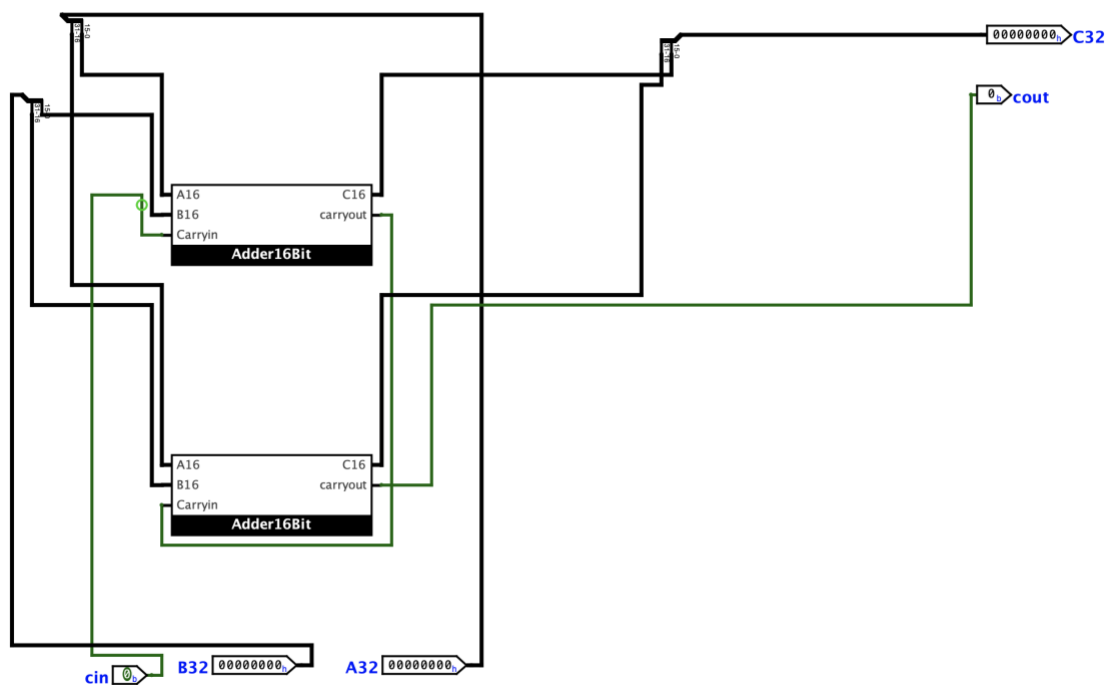
The implementation uses a series of five multiplexer stages, each corresponding to a shift of 1, 2, 4, 8, or 16 bits. Each stage is controlled by a specific bit of Sa, allowing shifts to be applied individually or combined to produce the desired total shift amount.

[illegible]

The overflow output is determined using logic based on the most significant bits of the inputs and result. Specifically, it checks whether the signs of the operands and the result indicate a signed overflow condition, ensuring that any incorrect result due to overflow is properly flagged.

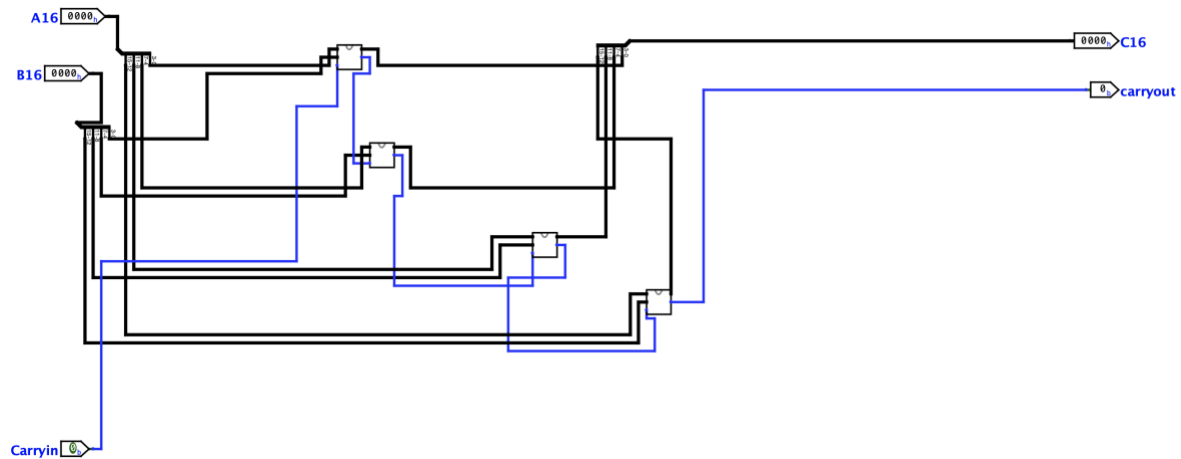
This part has the following sub systems:

32-bit adder:



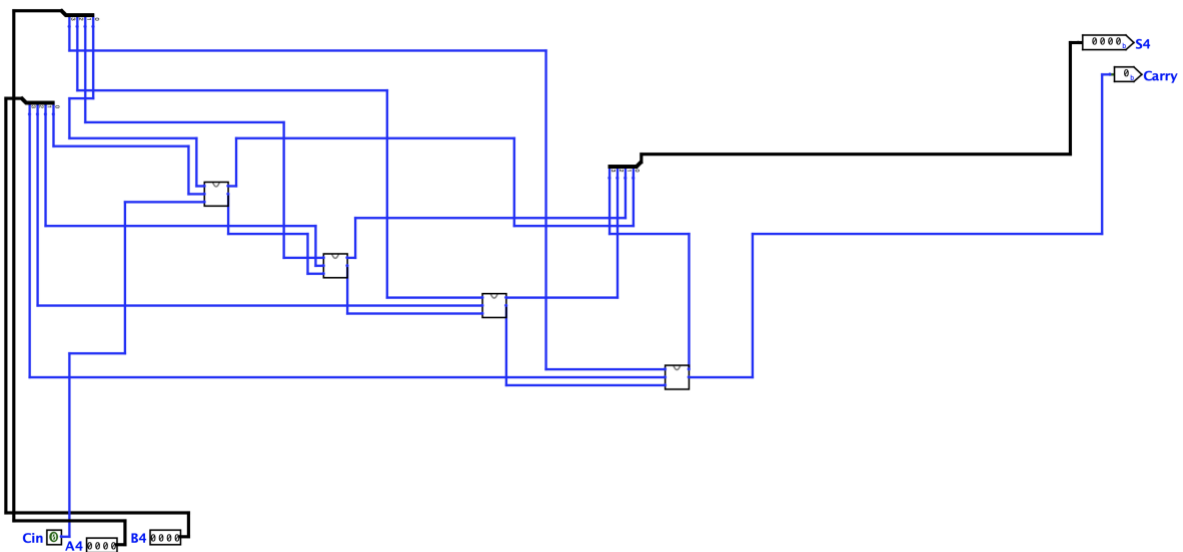
Performs 32-bit addition using inputs A, B, and Cin, and outputs the result along with an overflow signal when a signed overflow occurs.

16-bit adder:



Built from multiple 4-bit adders, this unit performs addition on 16-bit inputs while passing carry between blocks.

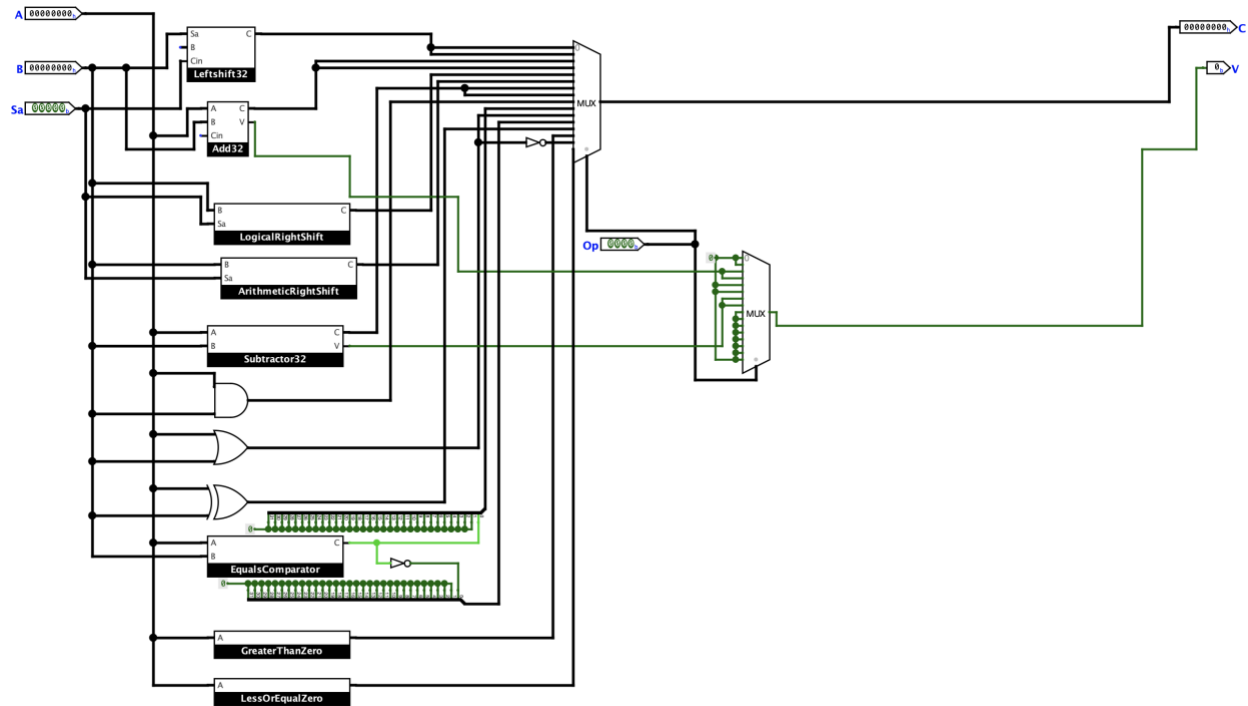
4-bit adder:



Combines four 1-bit full adders to perform addition on 4-bit values with carry propagation between each stage.

Part 3: ALU32

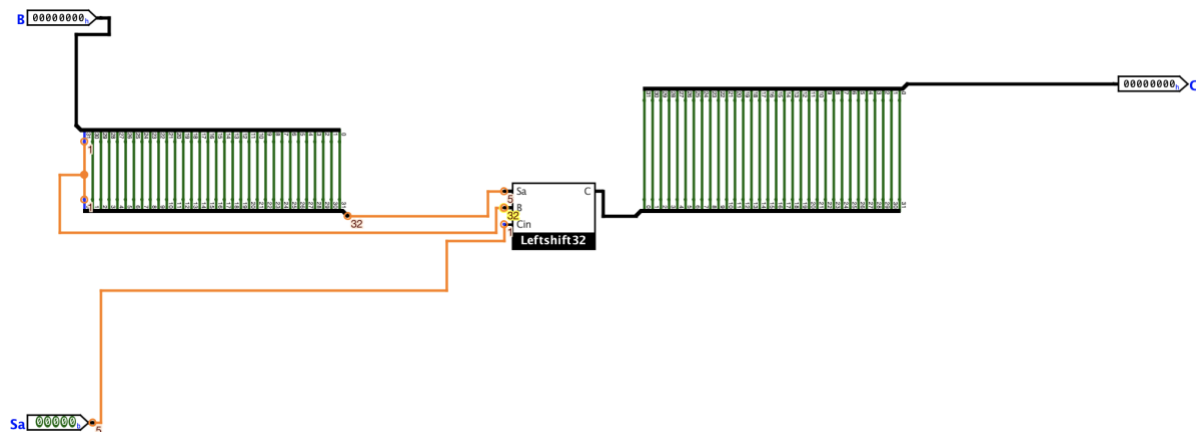
[illegible]



All operation results are connected to a multiplexer, which selects the final output based on Op. A separate control is used for the overflow output V, which is only active during addition and subtraction, and remains zero for all other operations. The design is built using several subcircuits, including the shifter, adder, subtractor, and comparison units.

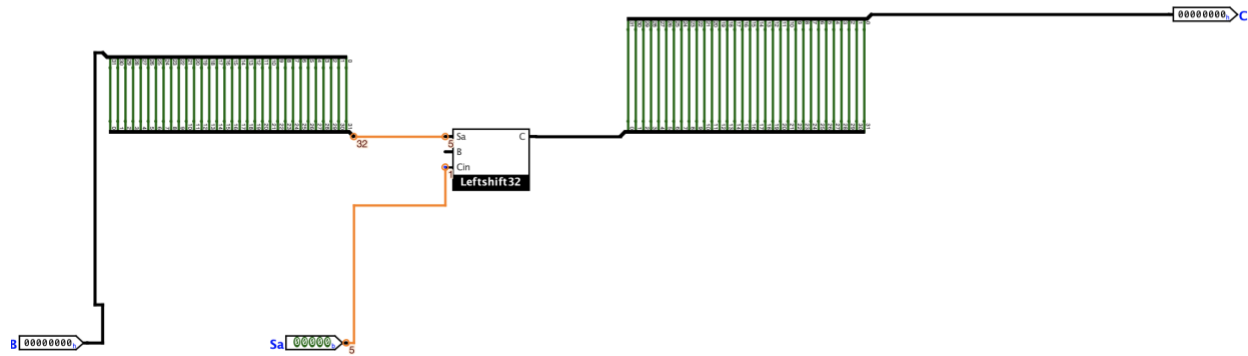
The ALU has the following subsystems

Arithmetic right:



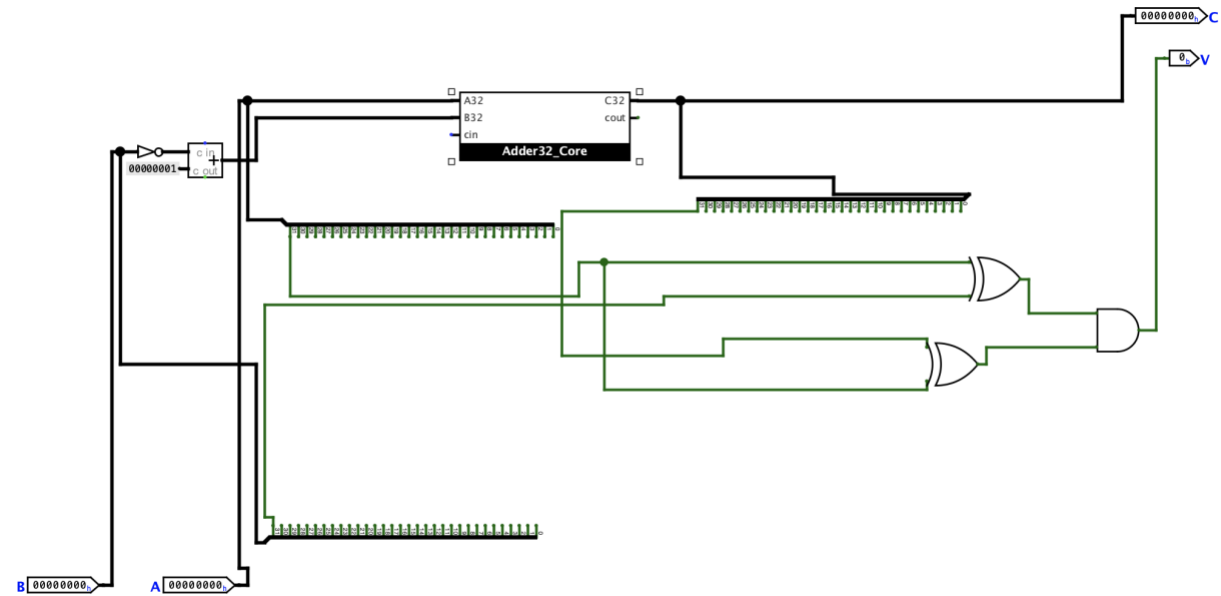
Shifts the input to the right while preserving the sign bit by filling the leftmost bits with the original MSB.

LogicalRightShift:



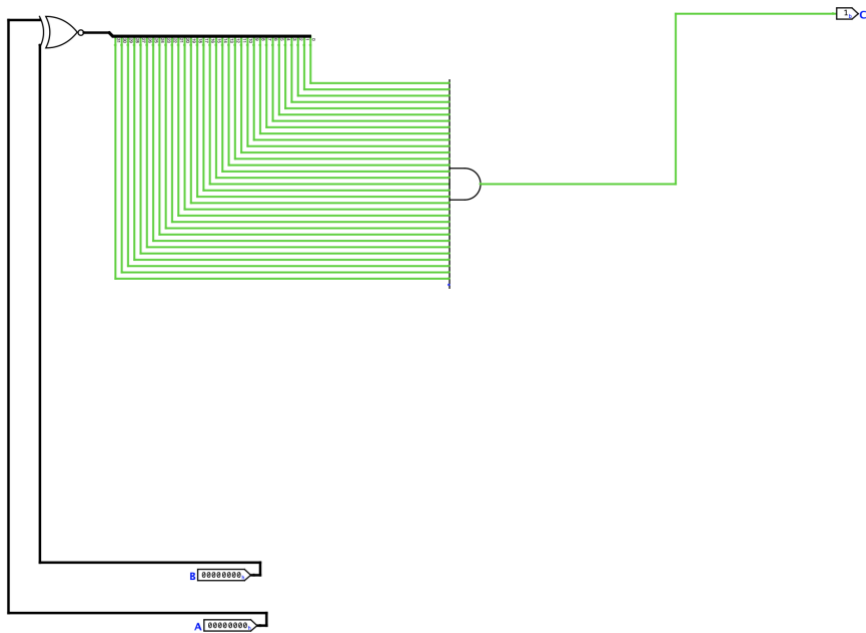
Shifts the input to the right and fills the leftmost bits with zeros.

Subtractor32:



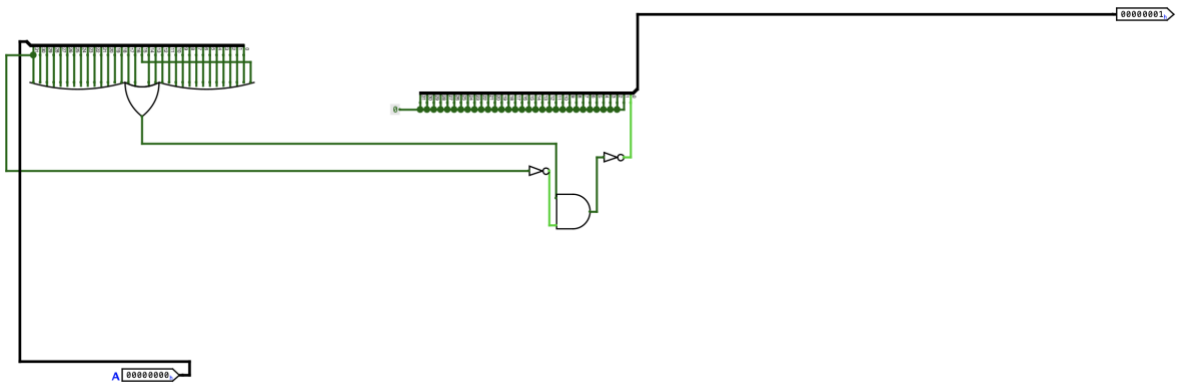
Computes the difference between A and B by using two's complement addition and produces a 32-bit result and overflow when needed.

EqualsComparator:



Compares A and B and outputs 1 if they are equal, otherwise outputs 0.

LessOrEqualsZero:



Checks if the input value is less than or equal to zero and outputs 1 if true, otherwise 0.

GreaterThanZero:

The circuit diagram for the `GreaterThanZero` function shows an 8-bit input `A` entering a 4-to-1 multiplexer. The multiplexer's output is connected to an 8-bit comparator. The comparator's output is an 8-bit signal that is high if the input `A` is greater than zero. The circuit uses a 4-to-1 multiplexer to select between the input `A` and its complement based on the MSB of `A`. The output is then compared to zero using an 8-bit comparator.

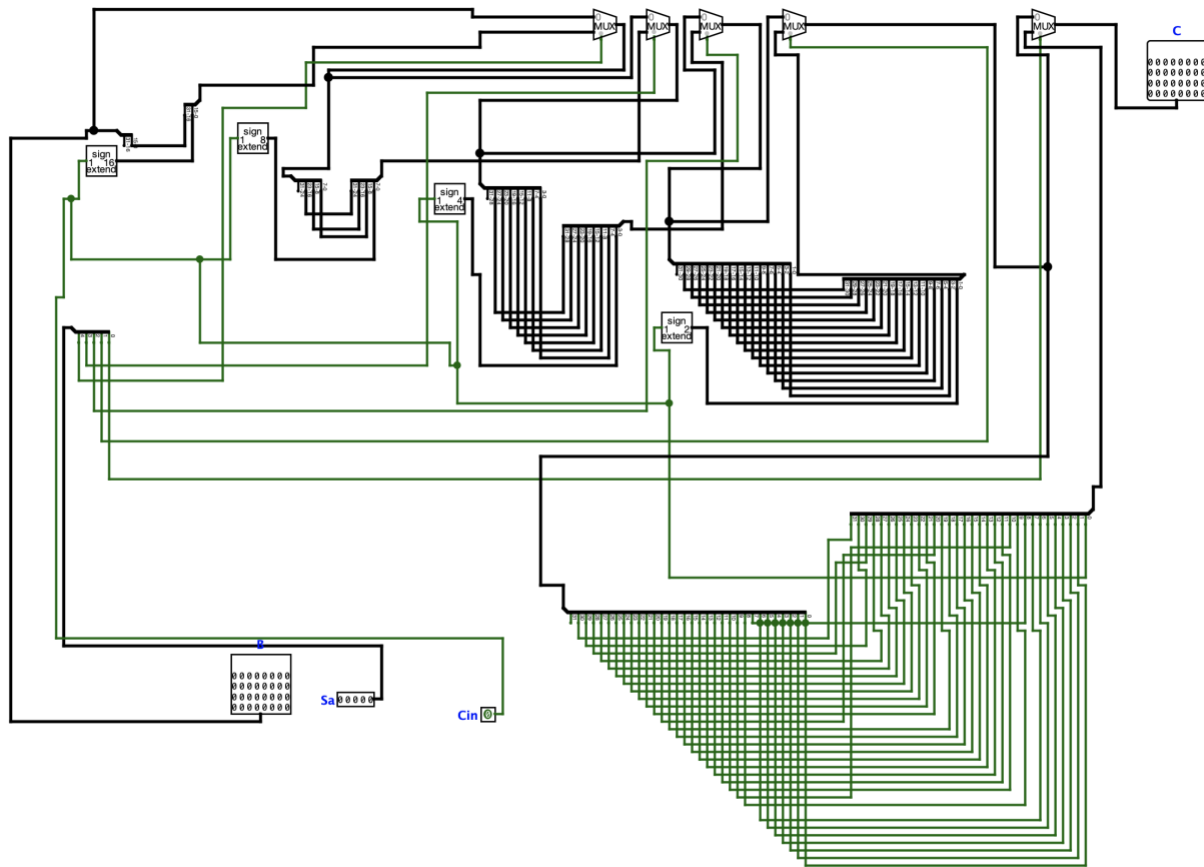
Checks if the input value is greater than zero and outputs 1 if true, otherwise 0.

Add32:

The diagram illustrates a 32-bit adder circuit. It features an **Adder32_Core** block with inputs **A32**, **B32**, and **cin**, and outputs **C32** and **cout**. The inputs **A** and **B** are 32-bit buses, each connected to a 32-bit bus indicator. The carry-in **Cin** is a single-bit input, also connected to a bus indicator. The carry-out **Cout** is a single-bit output, connected to a bus indicator. The circuit uses two 3-input OR gates and one 2-input AND gate to implement the carry logic. The first OR gate takes the carry-in **Cin** and the carry-out of the previous stage (which is **Cout** from the previous stage) as inputs. The second OR gate takes the carry-out of the previous stage and the carry-out of the current stage (which is **Cout** from the current stage) as inputs. The output of the first OR gate is connected to the carry-in of the current stage. The output of the second OR gate is connected to the carry-out of the current stage. The final carry-out **Cout** is connected to a bus indicator.

Performs 32-bit addition using inputs A, B, and Cin, and outputs the result along with an overflow signal when a signed overflow occurs.

LeftShift32:



Shifts the input value to the left by a specified amount and fills the empty bits with the value of Cin.

Critical Path of ALU:

The critical path of the ALU is mainly determined by the 32-bit adder, since it has the largest delay among all subcomponents. The adder is built from 32 one-bit adders connected in sequence, and each stage adds a small delay as the carry propagates through. This results in a relatively long delay compared to the other units. After the addition is completed, the result must also pass through a multiplexer before reaching the final output C[32], which adds one more level of delay. Therefore, the adder path plus the final multiplexer defines the overall critical path of the ALU.

Control Logic: ALU Operation Selection

The ALU uses the Op [4] input to determine which operation is selected.

Op [4]	Operation	C [32] Output	V Output
000x	Shift Left Logical	$B \ll S_a$	0
001x	Add	$A + B$	Overflow
0100	Shift Right Logical	$B \gg S_a$	0
0101	Shift Right Arithmetic	$B \ggg S_a$	0
011x	Subtract	$A - B$	Overflow
1000	AND	$A \& B$	0
1001	Equal	$(A == B) ? 1 : 0$	0
1010	OR	$A B$	0
1011	Not Equal	$(A != B) ? 1 : 0$	0
1100	XOR	$A \wedge B$	0
1101	Greater Than	$(A > B) ? 1 : 0$	0
1110	NOR	$\sim (A B)$	0
1111	Less or Equal	$(A \leq B) ? 1 : 0$	0

The opcode input Op [4] controls the multiplexer that selects the final ALU result. The overflow output V is only meaningful during addition and subtraction, and it is zero for all other operations.

Delay Estimate:

The estimated delay of the ALU is based on its longest path, which occurs through the 32-bit ripple-carry adder. Since the carry must propagate through all 32 one-bit stages, and each stage contributes about 2 gate delays, the adder delay is approximately 64 gate units. After that, the result must still pass through the final output multiplexer, adding 1 more unit of delay. Based on this, the total estimated critical path delay of the ALU is 65 gate delays.